

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	548795
Award Program	Workforce Development
Project Title	Innovations in Workforce Development: University Industry Consortium Fellows Program
Grant Start Date	09/01/2017
Grant End Date	10/31/2023
Total Grant Budget	\$5,507,057.78
Total FFAR Budget	\$2,753,528.89
Matching Funder(s)	Various
Final Report Date	02/06/2024
Project Director/Principal Investigator Information	
Full Name	John M Dole
Email	jmdole@ncsu.edu
Phone	919.623.2675
Grantee Organization Information	
Organization Name	NCSU
Address	2601 Wolf Village Way
City, State Zip	Raleigh, NC 27695
Tax ID	566000756

Authorized Signing Official (ASO) Information	
Name	Stefanie D. Saunders
ASO Title	Associate Director, Operations - Sponsored Programs
Email	sps@ncsu.edu
Phone	919-515-2444

General Information

- 1.1. *Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.*

Raleigh, North Carolina, 4th Congressional District (project office)

Locations of FFAR Fellows from the 2018 to the 2023 cohorts:

Athens, Georgia, 7th (Xiomy Pinchi, Emily Cason, Andrea Doblado, Sujan Paudel, Shreena Pradhan, Samantha Suber, Samuele Lamon, University of Georgia)

Berkeley, California, 13th (Delfina Grinspan, Dhruv Patel, Natalie Goh, Nicholas Karavolias, Alison Lui, University of California - Berkeley)

Bozeman, Montana, 1st (Mei Ling Wong, Montana State University)

Blacksburg, Virginia, 9th (Gambhir Lamsal, Virginia Tech)

Cleveland, Ohio, 11th (Gwen Donley, Abigail Roche, Case Western)

College Park, Maryland, 5th (Lovepreet Singh, Scott Cosseboom, Simon Sretenovic, University of Maryland)

College Station, Texas, 17th (Zachariah Vice, Sarah Kezar, Sarah Chu, Texas A&M University)

Columbus, Ohio, 3rd (Yamikani Ng'ona, Ohio State University)

Corvallis, Oregon, 4th (Lindsay Garcia, Oregon State University)

Davis, California, 3rd (Aoluwapemi Falade, Matt Davis, Mira Conyers, Maci Mueller, Krista Marshall, Dani Gelardi, Conor McCabe, University of California Davis)

East Lansing, Michigan, 8th (DeShae Dillard, Jaime Strickland, Morgan Mathison, Michigan State University)

Fargo, North Dakota, 13th (Abbeah Mae Navasca, North Dakota State University)

Fayetteville, Arkansas, 3rd (Enoch Ametepey, University of Arkansas)

Ft. Collins, Colorado, 2nd (Erica Patterson, Erin Jackson, Eva Keohane, Abby Barker, Maria Chavez, Siwook Hwang, Aaron Prairie, Ellie Ellis, Tyler, Thomas, Colorado State University)

Gainesville, Florida, 3rd (Paul Adunola, Federico Tarnonsky, Lushan Ghimire, Francesco Cappai, Felipe Alves Correa Carvalho da Silva, University of Florida)

Iowa City, Iowa, 2nd (Erika Ibarra-Garibay, Oluwatuyi Olowoyeye, Iowa State University)

Ithaca, New York, 23rd (Marjorie Hanneman, Haowen Hu, Gwyn Frederick, Lidia



Komondy, Ananda Portela Fontoura, Maria Gannett, Shiang-Wan Chin, Nicolas Morales, Agustin Olivo, Cornell University)

Knoxville, Tennessee, 2nd (Elias Zuchelli, Katlyn Smith, Autumn McLaughlin, Ekramul Ehite, Swati Mishra, University of Tennessee)

Lincoln, Nebraska, 2nd (Nikee Shreestha, Annie Krueger, Nate Korth, University of Nebraska)

Lubbock, Texas, 83rd (Ethan Tripless, Texas Tech)

Madison, Wisconsin, 2nd (Shane Hansen, University of Wisconsin-Madison)

Manhattan, KS 1st (Nicole Kucherov, Miriam Martin, Addison Carroll, Kansas State University)

Newark, Delaware, 7th (Kevin Oluoch, University of Delaware)

Pasadena, California, 28th (Yunqing Wang, California Institute of Technology)

Pullman, Washington, 5th (Barakatullah Mohammadi, Madeline Desjardins, Stephen Onayemi, Riley Reed, Alex Batson, Renan Stefanini Lopes, Alexa McDaniel, Phoebe Unger, Aichatou Waziri, Washington State University)

Raleigh, North Carolina, 4th (Kairi Tanaka, Grace Vincent, Hector Fajarado, Destiny Tyson, Connor Balfany, Vinicius Castelli Garnica, Heeduck Oh, Kevin Orta, Mariana Prieto, Alison Deviney, Lindsey Becker, Camilo Parada, Karlinton Flores, Lauren Anderson, Echo Pan, Daniela Pezzini, Xiaonan Shi, Sihui Ni, Jabeen Ahmad, Randi Butler, Shana McDowell, Enrique Pena, DeShae Dillard, Rachel Hammer, North Carolina State University)

Rutgers, New Jersey, 6th (Emma Waltman, Rutgers University)

Riverside, California, 41st (Danielle Stevenson, University of California – Riverside)

Saskatoon, Saskatchewan (Suneru Ussetti Mohottalalage, Chantel Chizen, University of Saskatchewan)

Starkville, Mississippi, 3rd (Graham Oakley, University of Mississippi)

State College, Pennsylvania, 5th (Chelsea Newbold, Olanrewaju Shittu, Zach Dashner, Sarah Bennett, Jeremy Sutherland, Pennsylvania State University)

St. Paul, Minnesota, 4th (Katrina Klett, Kelsey Peterson, Manuel Sabbagh, University of Minnesota)

St Louis, Missouri, 1st (Lauren Whitt, Caleb Grohman, University of Missouri)

Tuskegee, Alabama, 3rd (Inocent Ritte, Jamila Mweta, Tuskegee University)

Tucson, Arizona, 2nd (Shelby Hoglund, University of Arizona)

Urbana-Champaign, Illinois, 13th (Robert Bruhn, Seldon Kwafo, Jiayang Xie, Annemarie Mysonhimer, Sara Tondini, University of Illinois)

West Lafayette, Indiana, 4th (Rose Mumbi, Sujata Bogati, Ruth Eunice Centeno Martinez, Linda Beckett, Jason Graham, Paul Oladele, Purdue University)

Vancouver, Canada (Jonathan Beutler, University of British Columbia – Vancouver)

1.2. *How many new jobs were created because of this grant funding?*

1.5 (program management at NCSU)

1.3. *How many jobs were maintained because of this grant funding?*



1.2 (program management at NCSU)

- 1.4. *Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.*

No changes

- 1.5. *Has your organization undergone a recent name change? If so, please provide the new name of your organization.*

No, our organization has the same name.

2. Scientific Report

- 2.1. *Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.*

The Foundation for Food and Agriculture Research (FFAR) Fellows Program prepares the next generation of leaders in food and agriculture with a 3-year cohort-based training model for 30 students annually selected from among PhD students at U.S. and Canadian universities. The objectives of the FFAR Fellows Program are to: (1) Develop leadership competencies that enhance current and future individual productivity and well-being and enhance Fellows' capacity to contribute to the public good, (2) Connect young scientists across research domains and geographic areas to promote multi-disciplinary understanding and problem-solving, and (3) Broaden students understanding of their career options and create links to sector beyond academic (e.g., industry, government, NGO). These objectives are designed to meet the needs of the U.S. food and agriculture system for a career-ready STEM workforce with the capacity to work collaboratively across disciplines and sectors. In the first five years of the program (2017-2022) 110 students from 30 universities participated in 150 hours of leadership and professional development training in annual in-person sessions and monthly virtual sessions. These leadership competencies are stressed: (1) self-awareness and emotional self-management (2) time management and prioritization (3) interpersonal skills to work productively with others (4) mentoring and supervising (5) leadership through positive influence and (6) presenting scientific information to lay audiences. Fellows develop

close peer relationships within and across cohorts as they develop their skills and enter the workforce. Further network development is prompted by the program requirement to have multiple mentors, one of whom must be from industry. Participant selection is based primarily on the quality of the student and their potential to gain from leadership and professional development training and mentorship. Applicants have research areas in alignment with FFAR's Research Strategy, commitment to research that serves the public interest, and evidence of the desire to work with others to solve challenges in the food and agricultural sectors. Fellows have been financially sponsored by 43 companies, 14 commodity commissions, and nine foundations. Demographically, 35% are black or Latino. Forty-two students have graduated from their PhD program and moved into the workforce with the following breakdown among sectors: 48% into industry, 35% in academia, 10% in NGOs or non-profits, and 7% in government.

- 2.2. *Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).*

The FFAR Fellows Program was initially conceived during meetings of the University Industry Consortium, a 30-year old interdisciplinary private-public partnership with nearly fifty members (university, industry, and government partners). North Carolina State University took the lead to identify the structure and aims, and produce the ultimate proposal submitted to FFAR to create a multifaceted next generation STEM graduate training model to provide graduate students with real-world professional development experiences, integrating soft skills training into the graduate experience. Over the three-year cohort-based program, PhDs from multiple U.S. and Canadian universities and running the gambit of research areas (digital agriculture, bioinformatics, systems biology, soil microbiome, ruminant nutrition, etc.) engage in training to fill soft-skill non-disciplinary education gaps that currently exist in higher education. A 2022 third-party evaluation of the program (attached) documented the positive effects this training model has had on participants.

The impact of the program is magnified by the fact that the graduate students share what they have learned, promoting professional and leadership development within their labs, departments, and professional societies. Fellows give workshops on time management and prioritization, presenting science to lay audiences, and improving interpersonal communications and teamwork; they create career-exploration and networking opportunities for their peers by organizing industry panels and other interactions; and they practice "leadership from all levels" as team members, team leaders, and mentors and supervisors in their labs, departments and professional societies.



A key finding of the project has been the capacity for graduate students themselves to take ownership of their graduate school experience with confidence and to enhance leadership and professional development among their peers. This dissemination effect is addressing both the workforce needs in food and agriculture, by graduating career-ready PhD scientists capable of working with others to address challenges and opportunities in the food and agriculture system and addressing calls (from the National Academies of Sciences, Engineering, Medicine and others) to enhance STEM graduate training to include leadership, communication, and other professional competencies.

2.3. *Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:*

- a) *potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;*
- b) *intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and*
- c) *end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.*

Through their projects, the FFAR Fellows are impacting research in a myriad of ways. Their projects span the breadth of food and agricultural subjects from the soil to the plate, as well as communications, data science, financial, stakeholder well-being, and other topics. Most of the fellows are sponsored by companies or industry associations, which in combination with the academic component ensures that their work not only contributes to the science but also to the food and agricultural economy. The training they are receiving in the FFAR Fellows is advancing their productivity by helping them increase their self-awareness and emotional self-management, manage their time management effectively, improve their interpersonal skills to work productively with others, become leaders, and better communicate with others. The Fellows are at the beginning of their careers and the research impacts they generate are only starting.

2.4. *What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion*



dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The specific objectives of the original project and this award were to create and pilot a professional development program for graduate students with the goal of preparing a STEM-ready workforce for the food and agricultural sector. We used ongoing formative program evaluation to identify student needs, test assumptions, and create and fine-tune all program processes. We have been successful in defining and refining the student application and selection process; creating and executing a comprehensive 3-year training program; and preparing Fellows for industry, government, and non-profit careers. The objectives related to these goals are discussed in section 2.6.

- 2.5. *Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)*

No major changes.

- 2.6. *What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)*

Major activities: An advisory and selection group comprised of industry, FFAR and academic members reviews the applicants each spring. Applications are solicited for two categories: (1) Professional Development Program + Research Funding (Stipend/Professional Development category) and (2) Professional Development Program. Eight slots in the cohort are funded for the Stipend/Professional Development category. The remaining number of slots are designated for non-stipend fellows. All fellows participate equally in the Professional Development Program. Fellows attend four in-person sessions with two week-long sessions at North Carolina State University, a 4-day session in Washington, D.C., and a fourth 4-day session in St. Louis, MO. These are complemented by 10 webinars annually. The following leadership competencies are stressed: (1) self-awareness and emotional self-management (2) time management and prioritization (3) interpersonal skills to work



productively with others (4) mentoring and supervising (5) leadership through positive influence and (6) presenting scientific information to lay audiences. Fellows are also exposed to the role of government in scientific research and the policy-making process and to a wide-array of career paths. Fellows develop close peer relationships within and across cohorts as they develop their skills and enter the workforce. Further network development is prompted by the program requirement to have multiple mentors, one of whom must be from industry. The Professional Development Training Plan, attached, outlines the specific topics covered during the Fellows program.

Specific objectives: The objectives of the FFAR Fellows Program are to: (1) Develop leadership competencies that enhance current and future individual productivity and well-being and enhance Fellows' capacity to contribute to the public good, (2) Connect young scientists across research domains and geographic areas to promote multi-disciplinary understanding and problem-solving, and (3) Broaden students' understanding of their career options and create links to sector beyond academic (e.g., industry, government, NGO). These objectives are designed to meet the needs of the U.S. food and agriculture system for a career-ready STEM workforce with the capacity to work collaboratively across disciplines and sectors.

Results and conclusions: In the first five years of the program (2017-2022) 110 students from 30 universities participated in 150 hours of leadership and professional development training in annual in-person sessions and monthly virtual sessions. Participant selection is based primarily on the quality of the student and their potential to gain from leadership and professional development training and mentorship. Applicants have research areas in alignment with FFAR's Research Strategy, commitment to research that serves the public interest, and evidence of the desire to work with others to solve challenges in the food and agricultural sectors. Fellows have been financially sponsored by 43 companies, 14 commodity commissions, and nine foundations. Demographically, 35% are black or Latino. Forty-two students have graduated from their PhD program and moved into the workforce with the following breakdown among sectors: 48% into industry, 35% in academia, 10% in NGOs or non-profits, and 7% in government.

- 2.7. *Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.*

As a workforce development program focused on soft-skill training, the FFAR Fellows program gathered feedback information from fellows on all training components and used this information to fine-tune and finalize the 3-year training program. A third-party evaluator was engaged in 2023 to interview graduates of the program (report attached). Findings from this report are being used to inform a subsequent renewal to train an

additional three cohorts of fellows.

2.8. *List key stakeholders that could be served by results of this project.*

The direct beneficiaries from the FFAR Fellows Program are the Fellows themselves. Three cohorts of fellows were funded under the current award, with an additional three cohorts funded under a renewal. Indirectly the fellows' peers in their home departments are also beneficiaries in that most fellows bring back professional development knowledge from their Fellows training to the departments, specifically their lab groups, as well as their professional societies. The Fellows advisors benefit by the fact that graduate students supported by peers and mentors are more productive and more likely to complete their PhD. Academic departments benefit in that resources are used more efficiently when graduate students are successful and timely in their completion. Students, faculty, and members of the Fellow's mentor team benefit through the networked connections that the FFAR Fellows' program encourages across companies, universities, foundations, and other entities. One result is the greater attention given to issues of food and agriculture, leading to a virtuous feedback loop that increases interactions among these entities; involvement and training of exceptional students in food and agricultural sciences; and movement of graduating students into positions of influence and leadership.

2.9. *How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)*

Nothing to report (beyond dissemination of information through websites etc.; scholarly outputs are listed below, 3.2)



- 2.10. *Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)*

None.

- 2.11. *What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)*

The program focuses on providing professional development training for PhD students, empowering them to share their science and work with others to solve pressing problems in food and agriculture. Training for each cohort is based on in-person and virtual meetings. Based on formative evaluation in the first five years of the program, we now have a 3-year training program that includes 4 in-person meetings for each cohort and 30 webinars over the 3-year period. Professional Development Plans are also created annually by each Fellow in conjunction with the program's director and the fellows' mentors. PDP plans are shared with each fellow's mentor team through an on-line portal for review and comment and final versions of these plans are circulated to mentors. Fellows then reflect on these plans (documented in the portal in January and June) during the year and use these as a basis for discussions with their mentors. An additional dimension for professional development has emerged as fellows have begun to graduate from the program and become available to serve as formal or informal mentors, and to network with current fellows at FFAR Fellows events and scientific conferences.

- 2.12. *Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)*

In this reporting period, active cohorts included a total of 110 PhD students. The total number of fellows involved in the program (6 total cohorts) is 140, with 16 fellows in the first cohort rising to 30 in the most recent cohort.

3. Information Products

- 3.1. *Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.*

The peer-reviewed publications associated with Stipend+Professional Development fellows are listed in 3.2.

- 3.2. *Please provide a list of citations for the information products produced during the project.*

Published Peer-Reviewed Articles from the Stipend + Professional Development FFAR Fellows, Cohort 1 (2018-2021), Cohort 2 (2019-2022), and Cohort 3 (2020-2023)

Dr. Alexander M. Batson

Washington State University

Plant Pathology

Dissertation: The genetic basis of pathogenicity to spinach of *Fusarium oxysporum* f. sp. *spinaciae* (advised by Dr. Lindsey du Toit)

Graduation: 2022

2023 Batson, A. M., Woodhall, J. W., and du Toit, L. J. 2023. Real-time PCR assays for races of the spinach *Fusarium* wilt pathogen, *Fusarium oxysporum* f. sp. *spinaciae*. *Plant Dis.* 107:2633-2642.

2022 Batson, A. M., Gyawali, S., and du Toit, L. J. 2022. Shedding light on the spinach *Fusarium* wilt pathogen, *Fusarium oxysporum* f. sp. *spinaciae*. *Phytopathology* 112:2138-2150.

2022 Batson, A. M., Spawton, K. A., Katz, R., and du Toit, L. J. 2022. Cladosporium leaf spot, caused by *Cladosporium variabile*, in winter high tunnel production of spinach (*Spinacia oleracea*) in Maine, United States. *Plant Dis.* 106:2260.

2021 Batson, A. M., Fokkens, L., Rep, M., and du Toit, L. J. 2021. Putative effector genes distinguish two pathogenicity groups of *Fusarium oxysporum* f. sp. *spinaciae*. *Mol. Plant Microbe Interact.* 34:141-156.

Dr. Lindsey E. Becker

North Carolina State University

Plant Pathology

Dissertation: Unlocking the Wheat Seed Mycobiome: an Investigation of Drought Stress on Successive

Generations (advised by Dr. Marc Cubeta)
Graduation: 2023

Becker L. E. and Cubeta M.A. 2020. Increase of flower production and plant weight of *Calibrachoa x hybrida* by the soil fungus *Mortierella elongata*. Journal of Environmental Horticulture DOI: 10.24266/0738-2898-38.4.114

Becker L. E. and Cubeta M. 2020. Evaluation of calibrachoa cultivars for black root rot resistance, 2019. Plant Disease Management Reports. Vol 14: OT013

Dr. Scott Cosseboom

University of Maryland-College Park
Plant Pathologists

Dissertation: Diversity and pathogenicity of fungi causing fruit rot of wine grapes in the Mid-Atlantic (advised by Dr. Mengjun Hu)
Graduation: 2022

Cosseboom, S. D., and Hu, M. J. 2023. Utilizing disease prediction models to time fungicide applications for controlling ripe rot, caused by *Colletotrichum* spp., in Maryland vineyards. Plant Dis. doi.org/10.1094/PDIS-10-23-2114-RE.

Cosseboom, S. D., and Hu, M. J. 2023. Identification and pathogenicity of *Cladosporium*, *Fusarium*, and *Diaporthes* spp. associated with late season bunch rots of grape. Plant Dis. doi.org/10.1094/PDIS-01-23-0146-SC.

Cosseboom, S. D. and Hu, M. J. 2022. Predicting ripe rot of grape, caused by *Colletotrichum fioriniae*, with leaf wetness, temperature, and the crop growth stage. Phytofrontiers. doi.org/10.1094/PHYTOFR-05-22-0060-R.

Cosseboom, S. D. and Hu, M. J. 2022. Off-target selection of resistance to azoxystrobin in *Aspergillus* species associated with grape late season rots. Pestic. Biochem. And Phys. doi.org/10.1016/j.pestbp.2022.105227.

Cosseboom, S. D. and Hu, M. J. 2022. Ontogenic susceptibility of grapevine clusters to ripe rot, caused by the *Colletotrichum acutatum* and *C. gloeosporioides* species complexes. Phytopathology. doi.org/10.1094/PHYTO-01-22-0004-R.

Cosseboom, S. D. and Hu, M. J. 2021. A SYBR Green qPCR method for detecting and quantifying spores of *Colletotrichum acutatum* and *C. gloeosporioides* complexes causing ripe rot of grape. Plant Health Prog. 23:65-71.

Cosseboom, S. D. and Hu, M. J. 2021. Identification and characterization of fungicide resistance in *Botrytis* populations from small fruit fields in the Mid-Atlantic United States. Plant Dis. 105:2366-2373.

Cosseboom, S. D. and Hu, M. J. 2021. Diversity, pathogenicity, and fungicide sensitivity of fungal



species associated with late-season rots of wine grape in the Mid-Atlantic United States. *Plant Dis.* 105:3101-3110.

Dr. Alison V. Deviney

North Carolina State University

Biological and Agricultural Engineering

Dissertation: Systems thinking for sustainable manure management (advised by Dr. John Classen)

Graduation: 2022

Deviney A., Classen J. and Bruce J. (2024) Pathways to sustainable transitions in a complex agricultural system: a case study of swine waste management in North Carolina. *Front. Sustain. Food Syst.* 7:1292326. <https://doi.org/10.3389/fsufs.2023.1292326>

Deviney A., Classen J., & Bruce J. (2023). A methodology for using a multilevel perspective framework to analyze complex systems. *Methodological Innovations*, 16(2), 123-137. <https://doi.org/10.1177/20597991231160280>

Bird K.T., Nichols V.A., Garay E.S., Nowatzke M., Essary C., Post K.K., Deviney A.V., Welles J.S., Alao M.O., Dorbu F.E., Classen J.J., Koziel J.A., & Cortus E.L. (2022). Means, motive, and opportunity: A method for understanding stakeholder agency within food-energy-water systems. *Elementa: Science of the Anthropocene*, 10 (1): 00066. <https://doi.org/10.1525/elementa.2021.00066>

Welles J.S., Soriano N.C.T., Dorbu F.E., Pereira G.M., Rubeck L.M., Timmermans E.L., Ndayambaje B., Deviney A.V., Classen J.J., Koziel J.A., & Cortus E.L. (2021). Socio-economic and governance conditions corresponding to change in animal agriculture: South Dakota case study. *Sustainability (Switzerland)*, 13(19):10682. <https://doi.org/10.3390/su131910682>

Deviney A., Classen J., Bruce J., & Sharara, M. (2021). Sustainable Swine Manure Management: A Tale of Two Agreements. *Sustainability*, 13(1):15. doi: 10.3390/su13010015

Dr. Ananda Portela Fontoura

Cornell University

Animal Science

Dissertation: Developing nutritional therapies to mitigate heat and metabolic stress in lactating dairy cows (advised by Dr. Joseph McFadden)

Graduation: 2022

Li G., Yu X., Fontoura A.B.P., Javaid A., Sainz de La Maza-Escola V., Salandy N.S., Fubini S.L., Grilli E., McFadden J.W., Duan J.E. Transcriptomic regulations of heat stress response in the liver of lactating dairy cows. *BMC Genomics* 24:410, 2023.

Fontoura A.B.P., Sainz de La Maza-Escola V., Richards A.T., Tate B.N., Van Amburgh M.E., Grilli E., McFadden J.W. Effects of dietary organic acid and pure botanical supplementation on growth performance and circulating measures of metabolic health in Holstein calves challenged by heat stress. *Journal of Dairy Science* 106:2904-2918, 2023.

Fontoura A.B.P., Javaid A., Sainz de La Maza-Escola V., Salandy N.S. Fubini S.L., Grilli E., McFadden J.W. Heat stress develops with increased total tract gut permeability and dietary organic acid and pure botanical supplementation partially restores lactation performance in Holstein dairy cows. *Journal of Dairy Science*, 105:7842-7860, 2022.

Dr. Maria A. Gannett
Cornell University
Horticulture

Dissertation: Modifying plant-microbial interactions with high carbon soil amendments as a weed management strategy (advised by Dr. Jennifer Kao-Kniffin)
Graduation: 2023

Garcia J, Gannett M, Wei L, Cheng L, Hu S, Sparks J, Giovannoni J, Kao-Kniffin J (2022) Selection pressure on the rhizosphere microbiome can alter nitrogen use efficiency and seed yield in *Brassica rapa*. *CommsBio*, 5:959.

King WL, Kaminsky LM, Gannett M, Thompson GL, Kao-Kniffin J, Bell TH (2021) Soil salinization accelerates microbiome stabilization in iterative selections for plant performance. *New Phytol*: 1-10.

Dr. Natalie S. Goh

University of California- Berkeley
Chemical and Biomolecular Engineering

Dissertation: Advancing Nanomaterial Usage in Agriculture: Engineering Nanomaterials Towards Nanomaterial-Mediated Plant Genetic Engineering (advised by Dr. Markita P. Landry)
Graduation: 2022

Rosenberg, D.J., Cunningham, F.J., Hubbard, J.D., Goh, N.S., Wang, J.W., Haymen, E., Hura, G.L., Landry, M.P., Pinals, R.L. Mapping the Morphology of DNA on Carbon Nanotube-Based Sensors in Solution using X-ray Scattering Interferometry. *Accepted at JACS* (2023)

Wang, J.W., Squire, H.J., Goh, N.S., Ni, H.M., Lien, E.S., Wong, C., Gonzalez-Grandio, E., Landry, M.P. Delivered complementation in planta (DCIP) enables measurement of peptide-mediated protein delivery efficiency in plants. *Communications Biology* (2023).

Jackson, C.T., Wang, J.W., Gonzalez-Grandio, E., Goh, N.S., Mun, J., Krishnan, S., Landry, M.P. Polymer-Conjugated Carbon Nanotubes for Biomolecule Loading. *ACS Nano* (2021).

Goh, N.S., Zhang, H.*, Wang, J.W., Pinals, R.L., Gonzalez-Grandio, E., Demirer, G.S., Butrus, S., Fakra, S.C., Flores, A.D.R., Zhai, R., Zhao, B., Park, S.-J., Landry, M.P. Nanoparticle Cellular Internalization is Not Required for RNA Delivery to Mature Plant Leaves. *Nature Nanotechnology* (2021).

Voke, E., Pinals, R.L., Goh, N.S., Landry, M.P. *In Planta* Nanosensors: Understanding Biocorona Formation for Functional Design. *ACS Sensors* (2021).

Zhang, H.*, Cao, Y.*, Xu, D., Goh, N.S., Demirer, G.S., Cestellos-Blanco, S., Chen, Y., Landry, M.P., Yang, P. Gold- Nanocluster-Mediated Delivery of siRNA to Intact Plant Cells for Efficient Gene Knockdown. *Nano Letters* (2021).

Goh, N.S.*, Wang, J.W.*, Cunningham, F.J.*, Boozarpour, N.N., Pham, M., Landry, M.P. Nanoparticles for proteindelivery in planta. *Current Opinion in Plant Biology* (2021).

Dr. Nate Korth

University of Nebraska-Lincoln
Complex Biosystems

Dissertation: Identification and characterization of human gut microbiome active traits in cereal grains. (advised by Dr. Andrew Benson and Dr. James Schnable)

Graduation: 2022

Korth, N., Yang, Q., Van Haute, M.J., Tross, M.C., Peng, B., Shrestha, N., Zwiener, M., Mural, R.V., Schnable, J.C. and Benson, A.K., 2023. Grain Utilization by the Gut Microbiome as a Human Health Phenotype to Identify Multiple Effect Loci in Genome-Wide Association Studies of *Sorghum bicolor*. *bioRxiv*, pp.2023-09.

Yang, Q., Van Haute, M., Korth, N., Sattler, S., Rose, D., Juritsch, A., Shao, J., Beede, K., Schmaltz, R., Price, J., Toy, J., Ramer-Tait, A. E., and Benson, A.K., 2023. The waxy mutation in sorghum and other cereal grains reshapes the gut microbiome by reducing levels of multiple beneficial species. *Gut Microbes*, 15(1), p.2178799.

Gomes-Neto*, J.C., Pavlovikj*, N., Korth*, N., Naberhaus, S.A., Arruda, B. Benson, A.K., and Kreuder, A.J., 2023. Salmonella enterica induces biogeography-specific changes in the gut microbiome of pigs. *Frontiers in Veterinary Science*, 10, p.1186554. (*Co-first authorship)

Korth, N., Parsons, L., Van Haute, M., Yang, Q., Hurst, P., Schnable, J.C., Holding, D.R. and Benson, A.K., 2022. The unique seed protein composition of quality protein popcorn promotes growth of beneficial bacteria from the human gut microbiome. *Frontiers in microbiology*, p.2305.

Yang, Q., Van Haute, M., Korth, N., Sattler, S.E., Toy, J., Rose, D.J., Schnable, J.C. and Benson, A.K., 2022. Genetic analysis of seed traits in Sorghum bicolor that affect the human gut microbiome. *Nature Communications*, 13(1), pp.1-15.

Dr. Miriam Martin

Kansas State University
College of Veterinary Medicine

Dissertation: Investigation of Pain, Stress and Animal Welfare in Beef Cattle (advised by Dr. Johann Coetzee)

Graduation: 2021

Martin, M. S., M. D. Kleinhenz, A. V. Viscardi, A. K. Curtis, B. T. Johnson, S. R. Montgomery, M. E. Lou, and J. F. Coetzee. 2022. Effect of bupivacaine liposome suspension administered as a cornual nerve block on indicators of pain and distress during and after cautery dehorning in dairy calves. *Journal of Dairy Science*.

Martin, M.S., Kleinhenz, M.D, Viscardi, A.V, Curtis, A.K, Johnson, B.T, Montgomery, S.R., Lou, M.E., Coetzee, J.F. 2022. Effect of bupivacaine liposome suspension administered as a local anesthetic block on indicators of pain and distress during and after surgical castration in dairy calves. *Journal of Animal Science*

Martin, M.S., Kleinhenz, M.D., White, B.J., Johnson, B.T., Montgomery, S.R., Curtis, A.K., Weeder, M.M., Blasi, D.A., Almes, K.M., Amachawadi, R.G., Miesner, M.D., Baysinger, A.K., Nickell J.S., Coetzee, J.F. 2022. Assessment of pain associated with bovine respiratory disease and its mitigation with flunixin meglumine in cattle with induced bacterial pneumonia. *Journal of Animal Science*.

Martin, M.S., Kleinhenz, M.D., Montgomery, S.R. Blasi, D.A. Almes, K.M., Coetzee, J.F. 2022. Assessment of diagnostic accuracy of biomarkers to assess lung consolidation in calves with induced bacterial pneumonia using receiver operating characteristic (ROC) curves. *Journal of Animal Science*.

Martin, M., Kleinhenz, M.D., Schwartzkopf-Genswein, K.S., Melendez, D., Marti, S., Pajor, E.A., Janzen, E.D. and Coetzee, J., 2022. Characterizing the diagnostic sensitivity and specificity of pain biomarker in cattle using receiver operating characteristic curves. *Journal of Dairy Science*.

Martin, M.S., Kleinhenz, M.D., Montgomery, S.R., Cull, C., Seagren, J., Lechtenberg, K.F., Coetzee, J.F. 2022. Comparison of lidocaine administered alone or in combination with ethanol or bupivacaine liposomal suspension cornual nerve blocks or meloxicam to extend the duration of analgesia after scoop dehorning in male and female Holstein calves. *JDS Communications*.

Dr. Suneru Perera

University of Saskatchewan

Plant Pathologists

Dissertation: Value added processing of Canaryseed (cereal) protein. (advised by Dr. Michael Nickerson)

Graduation: 2023

Perera SP, Hucl P, L'Hocine. L, Nickerson MT. Microstructure and distribution of oil, protein, and starch in different compartments of Canaryseed (*Phalaris canariensis* L.). *Cereal Chem.* 2020; 1-18.

Perera S. P., Konieczny, D., Ding, K., Hucl, P., L'Hocine, L., & Nickerson, M.T. Techno-functional and nutritional properties of full-bran and low-bran canaryseed flour, and the effect of solvent-de-oiling on the proteins of low-bran flour and isolates. *Cereal Chemistry*, 2022; 1-24.

Dr. Kelsey Peterson

University of Minnesota

Plant and Microbial Biology

Dissertation: Surveying natural genetic variation to assess existing pest and pathogen resistance in *Silphium integrifolium* (advised by Dr. Yaniv Brandvain)

Graduation: 2023

Cassetta, E., Peterson, K., Bever, J.D., Brandvain, Y., VanTassel, D., Lubin, T.K., Alexander, H.M., Byers, D.L., Schiffner, S. and Turner, K., Adaptation of pathogens to their local plant host, *Silphium integrifolium*, along a precipitation gradient." *Ecosphere* 14, no. 6 (2023): e4565.

Peterson, K., Cheremond, E., Brandvain, Y., Van Tassel, D., Murrell E. (2022) Weight Gain of *Spodoptera frugiperda* larvae (Lepidoptera: Noctuidae) on leaf and floral tissues of *Silphium integrifolium* (Asterales: Asteraceae) differs by plant genotype. *Environmental Entomology*, 2022, *nvab146*

Dr. Camilo Humberto Parada Rojas

North Carolina State University

Plant Pathology

Dissertation: Revealing the sweetpotato resistome (advised by Dr. Lina Quesada-Ocampo)

Graduation: 2023

Parada-Rojas C. H. and Quesada-Ocampo L. M. (2022) *Phytophthora capsici* populations are structured by geography, host, and fluopicolide sensitivity. *Phytopathology*.

Parada-Rojas C. H., Granke L. L., Naegele R. P., Hansen Z., Hausbeck M. K., Kousik S., McGrath M. T., Smart C., and Quesada-Ocampo L. M. (2021) A diagnostic guide of *Phytophthora capsici* infecting vegetable crops. *Plant Health Progress*: *Invited paper for Managing Stubborn Oomycetes Special Issue *Plant Health Progress* 22: 404-414

Parada-Rojas C. H., Pecota, K., Almeyda, C., Yencho, G. C., & Quesada-Ocampo, L. (2021) Sweetpotato root development influences susceptibility to black rot caused by the fungal pathogen *Ceratocystis fimbriata*. *Phytopathology* 111: 1660-1669

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

The peer-reviewed publications associated with Stipend+Professional Development fellows are listed in 3.2.

- 3.4. *Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.*

Two websites feature the FFAR Fellows Program:

<https://foundationfar.org/ffar-fellows/>

This short page grounds the FFAR Fellows Program as a funded initiative of FFAR, and links viewers to the main website maintained by the program:

<https://ffarfellows.org/>

This website provides details on the program structure and training program, and details of the recruitment process and requirements to apply. A gallery page for each cohort provides links from each member to a detailed and attractive 'story map' on their research and background. The fellows' area of study, their universities, their sponsors, and their email and social media information is shared. Links to the social media accounts are also provided.

- 3.5. *Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).*

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. *Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)*



The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: John M. Dole

PI Signature: [Handwritten Signature] Date: 02/07/2024

Authorized Signing Official (ASO) Name: Stefanie D. Saunders

ASO Signature: Stefanie D. Saunders Date: 02/07/2024

Stefanie D. Saunders, Associate
Director, Ops. - Sponsored Programs

Digitally signed by Stefanie D. Saunders,
Associate Director, Ops. - Sponsored Programs
Date: 2024.02.07 09:17:50 -0500